

Workshop 4

Welcome!

Find a group to work with & say hi!

ECO 101: Principles of Microeconomics

Workshop Overview

- Measuring the response
- Interpreting elasticity
- Predicting price or quantity

Measuring the response

How large is the change?

Why not “the price went up by \$1.50”?

- Absolute changes depend on the units and where you started
 - \$1.50 on a gallon of gas is enormous. \$1.50 on a used EV is nothing.
- Elasticity = $\frac{\% \text{ change in the quantity that responded}}{\% \text{ change in the one thing that moved}}$
- Unit-free: comparable across markets, time, etc.
- Notation: Δ for change

Discrete vs. continuous data

- Discrete data: Last week: five buyers, five sellers, demand/supply as a staircase.
- Continuous data: Five hundred buyers and the steps get smaller. The staircase becomes a line.
 - We can use slopes. Careful: slope is not the elasticity
- Choose the appropriate formula for the data provided:
 - midpoint method: discrete data
 - point method: continuous data

Formulas

- Discrete data: **mid-point** method

$$|\varepsilon_D| = \left| \frac{\% \Delta Q_D}{\% \Delta P} \right| = \left| \frac{\frac{\Delta Q_D}{\text{avg } Q}}{\frac{\Delta P}{\text{avg } P}} \right|$$

- Continuous data: **point** method

$$|\varepsilon_D| = \left| \frac{\% \Delta Q_D}{\% \Delta P} \right| = \left| \frac{\frac{\Delta Q_D}{Q}}{\frac{\Delta P}{P}} \right| = \left| \frac{\Delta Q_D}{\Delta P} \frac{P}{Q} \right|$$

See videos for why we use these formulas and how to use them

Classification

- $|\varepsilon| > 1$: Elastic
- $|\varepsilon| = 1$: Unit Elastic
- $|\varepsilon| < 1$: Inelastic

Extreme elasticities

- $|\varepsilon| = \infty$: Perfectly Elastic
- $|\varepsilon| = 0$: Perfectly Inelastic

Note: If the sign is informative, we keep it.

- We use absolute elasticity of demand because the elasticity of demand is always negative

Demand: Slope, elasticity and spending

Notation: Δ for change

- Demand: $Q_D = 120 - 2P$, $\varepsilon_D = \frac{\Delta Q_D}{\Delta P} \frac{P}{Q}$

At market price	Market quantity	Elasticity of demand	Total Spending
$P = 40$	$Q = \underline{\hspace{2cm}}$	$\varepsilon_D = \underline{\hspace{2cm}}$	$P * Q = \underline{\hspace{2cm}}$
$P = 30$	$Q = \underline{\hspace{2cm}}$	$\varepsilon_D = \underline{\hspace{2cm}}$	$P * Q = \underline{\hspace{2cm}}$
$P = 20$	$Q = \underline{\hspace{2cm}}$	$\varepsilon_D = \underline{\hspace{2cm}}$	$P * Q = \underline{\hspace{2cm}}$

- Fill in the table. What patterns do you notice?

Worksheet Q1 (counts for grades)

Put answer on worksheet

Summary

- Elasticity measures how quantity responds to a change in one factor, all else held fixed
- Along a *linear* demand curve, the unit elastic point is spending or revenue maximizing
 - Spending and revenue are the same number ($P \times Q$) seen from the buyer's side and the seller's side of the market
 - In a competitive market nobody chooses where on the curve we sit; the market does
 - Revenue-maximizing price is not the profit-maximizing one as revenue ignores costs

Interpreting elasticity

Why this response? Interpreting sign and size

Types: Demand and Supply

Movements along a curve

(Own) price demand elasticity

- Sign: Negative if downward sloping demand

$$\varepsilon_D = \frac{\% \Delta Q_D}{\% \Delta P}$$

Supply elasticity

- Sign: Positive if upward sloping supply

$$\varepsilon_S = \frac{\% \Delta Q_S}{\% \Delta P}$$

- If the sign is informative, we keep it. If always the same use the absolute value

Types: Other

Shifts of the curve

How would you interpret the sign?

Cross Price elasticity: Change in quantity demanded for Good X when the price of good Y increases

- Negative ε_{CP} : _____
- Positive ε_{CP} : _____

$$\varepsilon_{CP} = \frac{\% \Delta Q_X}{\% \Delta P_Y}$$

Income elasticity: Change in quantity demanded when income changes

- Positive ε_I : _____
- Negative ε_I : _____

$$\varepsilon_I = \frac{\% \Delta Q_D}{\% \Delta I}$$

What makes the response big or small?

- W3: Markets are aggregations of individuals making choices
- If we want to understand why quantities respond, we think through how and why choices change

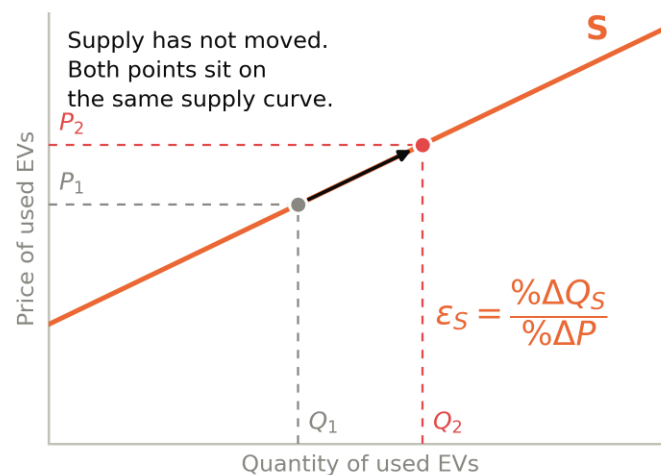
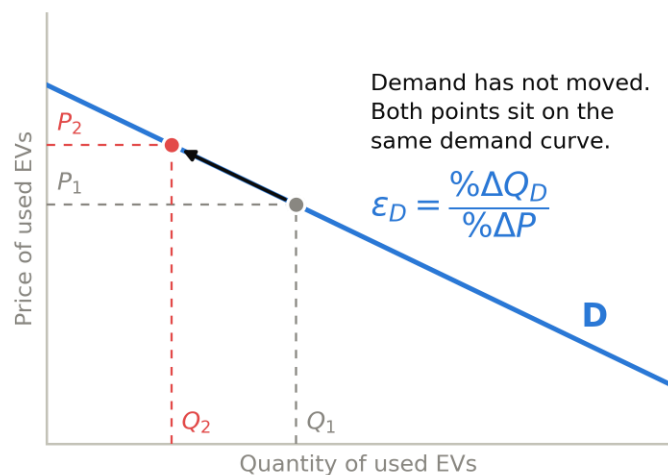
For example, when prices increase: are there other things I can buy? Do I want to produce more given my costs? How much time to adjust?

Predicting the price or quantity

The curve you move along is crucial

Used EV market

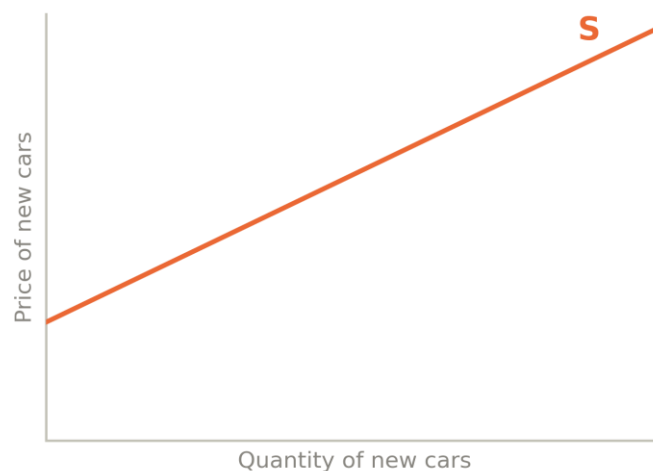
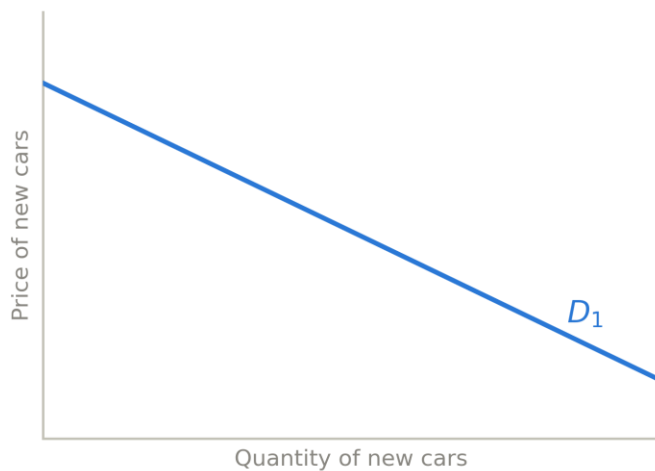
- W3: Price of gas increases => Prices for used EVs increased 2.1%
- Which elasticity to use? Recall:



New cars in Canada

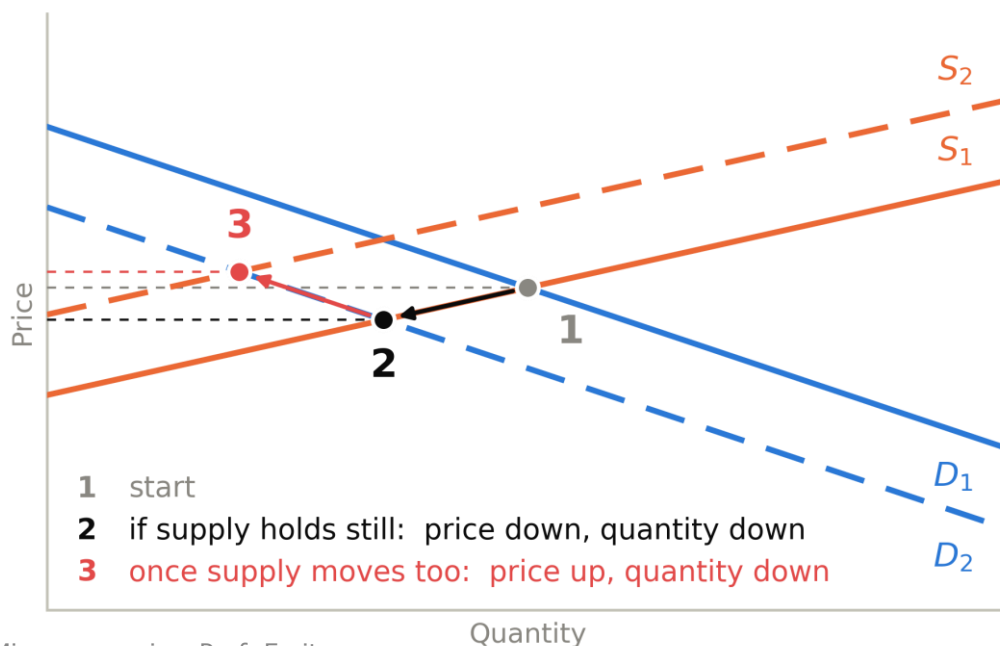
Only one change: lower incomes

- Assume new cars are a normal good
- Which elasticity to use to determine quantity change? Why?



What if both curves shift?

- W3: Pay attention to relative changes
- W4: Using elasticity of demand or supply requires a movement along the curve.
 - If the curve shifts you cannot use the original elasticities



Worksheet Q2 (counts for grades)

Put answer on worksheet

Summary

- Identify the reasons behind market changes to determine which elasticity to use to predict changes in prices, quantities, or revenue/spending
- When you want to use a given elasticity:
 - Make sure the denominator is the only factor changing, all else held fixed
 - Ask, why does the quantity in the numerator change? What determines the size of the change?
- If you want to interpret it on a demand/supply diagram, ask, does it cause a movement along the curve or of the curve? Why?

Something to think about at home

Don't forget to give me your
completed worksheets

See you next workshop!
